When Food Exists but Hunger Persists

When Food Exists but Hunger Persists: Identifying Food Supply–Access Disconnects in U.S. Counties

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Abstract

Food insecurity remains a persistent issue in the United States despite the country's substantial economic wealth and food production capacity. In 2023 alone, approximately 18 million U.S. households experienced food insecurity, highlighting a paradox in which food systems appear robust while many communities still lack reliable access to nutritious food. This research examines the structural mismatch between food infrastructure and actual food access at the county level across the United States. By integrating multiple national datasets, the study constructs two composite indices: the Food Infrastructure Index (FII), representing the density of grocery stores, supercenters, and SNAP-authorized retailers, and the Low Access Indicator (LAI), representing the proportion of low-income and low-access populations. These indices are standardized and used to calculate an Access Mismatch Score that identifies counties where food infrastructure appears sufficient but residents still face significant access barriers. Additional socioeconomic factors such as median household income and vehicle access are incorporated to distinguish between physical access limitations and broader economic disadvantage. The findings aim to reveal geographic patterns of food access inequality and highlight counties where food systems fail despite apparent supply. Ultimately, this research seeks to provide insights that can support policymakers, public health agencies, and community organizations in developing more targeted strategies to address food insecurity.

Data Acquisition

The data used in this study were obtained from several publicly available national datasets that provide information on food access, food infrastructure, and socioeconomic conditions across U.S. counties. The primary dataset used to measure food accessibility was the USDA Food Access Research Atlas (2019), which provides variables describing low-income and low-access populations in both urban and rural areas. To measure the availability of food infrastructure, the study utilizes the USDA Food Environment Atlas (2020), which contains data on the density of grocery stores, supercenters, and SNAP-authorized retailers across counties.

Additional socioeconomic and transportation-related variables were collected from the American Community Survey (ACS) 2015–2019, including median household income and the percentage of households without access to a vehicle. These variables provide important contextual information for distinguishing whether food access challenges are driven by infrastructure limitations or broader socioeconomic factors.

To further explore the role of retail food environments, the project also incorporates a Dollar General store location dataset provided by Pareto Software, which tracks the geographic distribution of these stores across the United States. This dataset allows the analysis to evaluate whether low-cost, low-nutrition food retailers are disproportionately concentrated in areas

experiencing structural food access mismatches. All datasets were integrated at the county level using Federal Information Processing Standards (FIPS) codes as the common geographic identifier. After merging the datasets, variables were cleaned, standardized, and prepared for analysis to support the construction of composite indices and subsequent spatial and statistical analysis.

DATA CLEANING AND PREPROCESSING

Website URL : https://when-food-exists-but-hunger-persists.base44.app

GitHubURL:
https://github.com/dnyanadab/DATA-MINING-PROJECT-GROUP-13.git

Dataset 1: USDA Food Access Research Atlas
This dataset is sourced from the USDA Food Access Research Atlas and provides county-level indicators related to food access across the United States. The dataset includes measures of population distance to grocery stores, income-related food access barriers, and low-access population counts.

Data Source
USDA Food Access Research Atlas
https://www.ers.usda.gov/data-products/food-access-research-atlas/

Research Relevance
Food access is a critical dimension of food security. Counties with limited access to grocery stores often rely on convenience retail, which may provide limited access to nutritious food. This dataset enables analysis of geographic disparities in food access across the United States.

1) **Handling issues/Noise in the data :** We identified significant missing data in variables related to distant population counts e.g (lalowi20, lanhopi20), with some columns missing up to 97.92% of their values. We made use of food_access.isnull().sum() to fill in these gaps.

2) **Understanding the data:** We used food_access.dtypes.value_counts() to get a summary of data types used. Numerical variables include PovertyRate and Pop2010. Categorical variables include State and Urban.

3) **Basic Statistical Analysis:** Summary statistics was performed specifically for lapop10share (share of population living far from a grocery store). This scopes the distribution of the primary outcome-proxy variable and highlights skewness or extreme values.

4) **Data Quality Assessment :** The dataset is highly comprehensive, containing 147 columns and 72,531 rows. It addresses the project's goal of analyzing geographic disparities in food access and identifying food scarcity.

5) **Advanced Data Understanding:** There is a heatmap showing correlation between variables like PovertyRate, MedianFamilyIncome, and LowAccessShare. County_low_income_low_access_10_share and county_low_access_10_share are highly correlated to each other. The KDE plot between Tract and County Share density show more variation for small values of tract_low_access_10_share.

Visualizations and Insights:

1) **Histplot :** Cleaned Tract-Level Low Access Share (10-mile)
   Insight : Low value share has the highest value. The hisplot is right skewed.
   As the share of low access increases, the number of tracts decreases, but it doesn't drop to zero.

2) **Boxplot :** Low Access Share by Urban Classification
   Insight : Rural areas have a significantly higher median low access share than urban areas. The interquartile range (IQR) for rural areas is higher than almost the distribution of the urban data.

3) **Heatmap :** County-Level Correlation Structure (Cleaned)
   Insight : The heatmap shows a strong positive correlation between PovertyRate and county_low_income_low_access_10_share.

4) **Scatterplot :** Poverty Rate vs Low Access Share (Cleaned)
   Insight : There is no clear, strong linear trend (upward or downward). This suggests that a high poverty rate does not automatically guarantee a high low-access share, and vice versa.

5) **Scatterplot :** County Poverty vs Low-Income Low-Access
   Insight : Low Access Share generally stays below 0.15 (15%). This indicates that for many U.S. counties, food access issues affect a relatively small portion of the total population, even when poverty is present.

6) **Scatterplot :** Log-Transformed Tract Low Access Share
   Insight : Low value share has the highest value. The hisplot is right skewed. We observe a rise in share value at 0.7

7) **Scatterplot :** Poverty Rate vs Low Access Share (Cleaned)
   Insight : There is no linear relationship between the variables. Most tracts in this dataset fall into a moderate category where poverty is relatively low and access to services (or food/resources) is relatively high.

8) **KDEPlot :** Tract vs County Share Density
   Both distributions are heavily **right-skewed**. The vast majority of both census tracts and counties have a very low share of low access populations, with the highest density of data points clustered near 0.0 to 0.1.

9) **Lorenz Curve :** Inequality in County level food access
   Insight : The curve becomes significantly steeper as it moves toward the right (the top 20% of counties).The lowest 20% of counties have almost none of the resource.

10) **Regression Plot :** PovertyRate vs county_low_access_10_share
   Insight : The line is downward-sloping regression line. This indicates a negative correlation between the two variables: As the PovertyRate increases, the county_low_access_10_share tends to decrease.

Before/After:

Before Cleaning : The raw dataset was 72,531 rows and 147 columns, representing census-tract level data across the U.S**.**

Dataset 1: Food Access Research Atlas (Raw Variables)

The dataset contains census tract–level indicators related to food access, socioeconomic conditions, and demographic characteristics. The variables can be grouped into several categories.

1. Geographic Identifiers

Column	Description
CensusTract	Unique census tract identifier
State	U.S. state name
County	County name
Urban	Indicator identifying whether the tract is urban or rural

These variables allow spatial identification and enable aggregation to county-level datasets.

After cleaning : The high-dimensional data was transformed by organizing the 147 variables into logical thematic groups to facilitate targeted analysis: Geographic Identifiers: Consolidating CensusTract, State, County, and Urban status to enable spatial aggregation. By identifying and grouping variables (Income, Poverty, Low Access Counts, Demographics), we improved usability of the data, reducing the noise from irrelevant columns. The cleaned data includes transformations from raw population counts to proportions (using the *share suffix), which standardizes measurements across tracts with varying population sizes.

Dataset 1: Processed Data Column Descriptions

Column Name	Description	Data Type
CountyFIPS	Standardized county identifier used to merge datasets	Identifier
State	State name	Categorical
CountyName	Cleaned county name used for visualization	Categorical
Pop2010	Total county population	Numerical
low_access_share	Percentage of county residents with limited food access	Numerical
access_z	Standardized score representing food access limitations	Numerical
infra_z	Standardized score representing food infrastructure availability	Numerical
MismatchScore	Difference between food access demand and available infrastructure	Numerical
HighMismatch	Indicator identifying counties where infrastructure fails to meet access needs	Binary
FoodSystemType	Classification of counties based on food access and infrastructure conditions	Categorical
InfrastructureEfficiency	Measure of how effectively food retail infrastructure serves the population	Numerical
RetailDiversity	Index measuring diversity of food retail establishments	Numerical
ExpectedAccess	Estimated food access level predicted by infrastructure	Numerical
ActualAccess	Observed food access level	Numerical
AccessGap	Difference between expected and actual food access	Numerical
SystemBalance	Measure of equilibrium between food demand and supply	Numerical

Dataset 02 : USDA Food Environment Atlas (FoodEnvironmentAtlas.xls)

Sheet used: STORES
Scope: 3,143 US counties (all 50 states + DC)
Granularity: County-level food retail infrastructure metrics

Research Relevance : This dataset is used to explore food access inequality across US counties.

We examine how the density of grocery stores, supercenters, convenience stores, SNAP retailers, and WIC outlets varies geographically — and construct a composite Food Infrastructure Index to quantify access disparities.

Data Source
USDA Economic Research Service – Food Environment Atlas
URL: https://www.ers.usda.gov/data-products/food-environment-atlas/

1) **Handling issues/Noise in the data :** Missing values for SNAPSPTH17 and WICSPTH16 were imputed using the state-level median. The state-level median was chosen instead of the national mean to account for regional variations and to be robust against outliers within specific states. FIPS codes (county identifiers) were initially read as integers. They were converted to 5-digit zero-padded strings (e.g., 1001 → '01001') to maintain the standard geographic format. Column names were cleaned using str.strip() to remove any accidental leading or trailing whitespace.

2) **Understanding the data:**
   Categorical data : State: Two-letter state abbreviations (e.g., "AL"). FIPS: Initially read as an integer, but we converted it later.

   Numerical data : Columns such as GROCPTH16 (Grocery stores per 1,000 pop), SUPERCPTH16 (Supercenters), CONVSPTH16 (Convenience stores), SNAPSPTH17 (SNAP retailers), and WICSPTH16 (WIC outlets).

   String data - County: The names of the counties (e.g., "Autauga").

   We have applied feature selection to drop over 30 "percent-change" (PCH_*) columns because they are derived features that would introduce multicollinearity in future modeling.

   The FIPS column is transformed from an integer type to a zero-padded 5-character string (e.g., 1001 → '01001') to meet standard formatting.

3) **Basic Statistical Analysis:** We have performed statistical summary on features such as :

   - GROCPTH16: Grocery stores per 1,000 population (2016)

   - SUPERCPTH16: Supercenters and club stores per 1,000 population (2016)

- CONVSPTH16: Convenience stores per 1,000 population (2016)
- SNAPSPTH17: SNAP-authorized retailers per 1,000 population (2017)
- WICSPTH16: WIC-authorized retailers per 1,000 population (2016)

These statistics reveal significant variability across counties, particularly in the density of SNAP-authorized retailers and convenience stores compared to grocery stores and supercenters. The dataset contains over 10 thematic sheets (e.g., ACCESS, STORES, RESTAURANTS). This analysis integrates metrics exclusively from the **STORES** sheet to focus on retail infrastructure. Data integration across counties is facilitated by the FIPS code. The notebook performs a critical data cleaning step by converting FIPS codes from integers to zero-padded 5-character strings (e.g., 1001 → '01001') to ensure compatibility with standard geographic datasets.

4) **Data Quality Assessment :** The dataset is highly usable for comparative analysis as it provides per-1,000-population density metrics (e.g., GROCPTH16 for grocery stores, SNAPSPTH17 for SNAP retailers). To improve usability and avoid multicollinearity, our team opted to retain only the most recent density columns and drop over 30 historical or percent-change columns. The dataset is directly aligned with the research goal of exploring food access inequality across the United States. It specifically enabled us to:
   i) Examine the geographic variation of grocery stores, supercenters, convenience stores, and assistance-authorized retailers (SNAP and WIC).

   ii) Construct a composite Food Infrastructure Index to quantify and compare access disparities across different counties and states.

   Aggregation Bias: Because the data is provided as county averages, it may mask significant intra-county variations, such as the differences between urban and rural sub-areas within the same county.

   Social and Geographic Bias: Density metrics (stores per 1,000 people) may under-represent Indigenous and rural communities. In these areas, population denominators are small, which can skew metrics even when geographic coverage is expansive.

5) **Advanced Data Understanding:** We have examined relationships between different store densities using a correlation heatmap. We also performed data consolidation by aggregating county-level data to the state level to calculate the mean FoodInfrastructureIndex per state**.**

Visualizations and Insights:

1) **Heatmap :** Correlation Matrix : Food Retail Density Variables
   Insight : SUPERCPTH -16 and CONVSPTH-16 have the lowest correlation amongst each other of -0.04. WICSPTH-16 and GROCPTH-16 have a correlation of 0.67.

2) **Boxplot :** Store density outlier check
   SNAPSPTH17 - This category has the highest median density and the widest Interquartile Range (IQR). SUPERCPTH16- This category has the lowest density and the tightest distribution.

3) **Histogram :** Distribution of Food Infrastructure Index Across Counties
   The data follows a right-skewed distribution. Most counties fall within the 0.2 to 0.5 range. The highest frequency of counties occurs between 0.3 and 0.4, indicating this is the base level of food infrastructure for the average county in this dataset.

4) **Scatterplot :** Raw Grocery vs Convenience Store Density
   There is no strong linear relationship between the two variables. An increase in grocery store density does not necessarily predict an increase (or decrease) in convenience store density.

5) **Choropleth plot :** Food Infrastructure Availability Across U.S. Counties
   In Alaska and parts of the West, you see bright yellow spots surrounded by deep purple. This illustrates a Hub-and-Spoke model. The bright hot spots in the Midwest (Nebraska, Kansas, Iowa) represent a high Location Quotient for food infrastructure.The deep purple clusters in the Southeast (Mississippi Delta) and Southwest (Navajo Nation regions) indicate a lack of Proximal Access.

6) **Scatterplot :** Healthy vs Convenience Food Retail Environment
   Insight : The highest density of data points is clustered in the lower-left quadrant. This means that the vast majority of counties or areas surveyed have:0.1 to 0.3 grocery stores per 1,000 people , 0.2 to 0.8 convenience stores per 1,000 people. This indicates that convenience stores are generally more prevalent per capita than grocery stores in most regions.

7) **Bar chart :** Counties with the Highest SNAP Retailer Density
   Insight : Yukon-Koyukuk and Lake and Peninsula have a density significantly higher (between 5 and 6) than the rest of the counties.

8) **Histogram :** Counties Where Convenience Stores Dominate Food Retail
   Insight : The vast majority of counties fall into the very first two bins, specifically between values of 0 and 10. The data is right-skewed.

9) **Linechart : Inequality in Food Infrastructure Across Counties**
   Insight : The majority of the graph exhibits a steady slope. Between county 500 and county 2,500, the index climbs from roughly 0.25 to 0.50.

10) **Barchart : Average Retail Food Infrastructure per County**
   Insight : Snap retailers has the highest average retail food infrastructure per county.Supercenter has the lowest average retail food infrastructure per county.

Before/After:

Before Cleaning: The dataset contained over 30 columns, many of which were redundant or outside the research scope (e.g., historical data from 2009 or specific percentage change metrics). There were significant NaN values, particularly in the WIC_STORES16 and PCH_WIC_08_16 columns, which could skew statistical analysis.

Raw data snapshot before cleaning

```
stores = pd.read_excel(
    FE_PATH,
    sheet_name="STORES"
)

stores.head()
```

	FIPS	State	County	GROC11	GROC16	PCH_GROC_11_16	GROCPTH11	GROCPTH16	PCH_GROCPTH_11_16	SUPERC11	...	PCH_SNAPS_12_17	SNAPSPTH12	SNAPSPTH17	PCH_SNAPSPTH_12_17
0	1001	AL	Autauga	5	3	-40.000000	0.090581	0.054271	-40.085748	1	...	19.376392	0.674004	0.804747	19.397900
1	1003	AL	Baldwin	27	29	7.407407	0.144746	0.139753	-3.449328	6	...	36.927711	0.725055	0.890836	22.864524
2	1005	AL	Barbour	6	4	-33.333333	0.219370	0.155195	-29.254287	0	...	3.349282	1.280590	1.424614	11.246689
3	1007	AL	Bibb	6	5	-16.666667	0.263794	0.220916	-16.254289	1	...	11.794872	0.719122	0.801423	11.444711
4	1009	AL	Blount	7	5	-28.571429	0.121608	0.086863	-28.571429	1	...	5.701754	0.657144	0.692374	5.361034

5 rows × 39 columns

After Cleaning : Retained only the essential geographic identifiers (FIPS, State, County) and the most recent counts (2014-2016) for retail types like Grocery Stores, Supercenters, Convenience Stores, and specialized outlets (SNAP/WIC). Missing values were handled via Zero Imputation (under the assumption that no record often implies zero specialized stores in rural counties) or Mean Imputation, ensuring a complete matrix for index calculation. A new composite metric, FoodInfrastructureIndex, was calculated.

cleaned dataset stats

```
stores_clean[[
    "GROCPTH16",
    "SUPERCPTH16",
    "CONVSPTH16",
    "SNAPSPTH17",
    "WICSPTH16"
]].describe()
```

	GROCPTH16	SUPERCPTH16	CONVSPTH16	SNAPSPTH17	WICSPTH16
count	3143.000000	3143.000000	3143.000000	3143.000000	3143.000000
mean	0.243835	0.018255	0.595571	0.917416	0.212930
std	0.216536	0.022460	0.304968	0.370847	0.189319
min	0.000000	0.000000	0.000000	0.040180	0.004938
25%	0.132983	0.000000	0.401678	0.680023	0.112934
50%	0.186116	0.014654	0.542005	0.861273	0.162162
75%	0.279887	0.029268	0.722071	1.101538	0.245756
max	3.267974	0.250752	3.680982	5.902454	3.262643

Dataset 03 : ACS 2019 datasets (DP 03, DP 04 , and DP 05)

Data Source & Method

The ACS 5-Year Estimates are published annually by the U.S. Census Bureau and are freely downloadable from data.census.gov. The three Data Profile tables (DP03, DP04, DP05) were downloaded at the county level for the 2019 survey year — the most recent pre-COVID full-year estimates available at time of collection.

Why ACS and not an API?

The Census Bureau does offer a public API (api.census.gov). However, bulk county-level extraction across three full DP tables is substantially simpler via direct CSV download, and the resulting files are identical in content.

Why 2019?

2019 is the last pre-COVID year, avoiding pandemic-driven distortions in employment, poverty, and housing metrics

Temporally aligned with the food access dataset (2015–2019 window)

ACS 5-year estimates pool 5 years of survey data for stable county-level estimates — appropriate for rural counties with small sample sizes

Columns Selected

From hundreds of ACS variables, we extract the six most relevant to food access research:

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ACS Variable	Renamed	Rationale
DP03_0062E	MedianIncome	Primary economic capacity indicator
DP03_0119PE	PovertyRate	Core food insecurity driver
DP03_0066PE	UnemploymentRate	Structural economic distress marker
DP04_0058PE	NoVehicleRate	Transportation barrier to food access
DP04_0003PE	HousingVacancyRate	Housing instability / community stability
DP05_0018E	MedianAge	Mobility and vulnerability indicator
DP05_0037PE	HispanicShare	Demographic equity variable

1) **Handling issues/Noise in the data :** The raw ACS data often uses symbols like (X) or ***** as placeholders for missing or non-applicable values.To handle these non-numeric strings, we made use of pd.to_numeric() with the errors='coerce' parameter. This automatically converts any non-numeric noise (like (X)) into NaN (Not a Number) values.After converting selected columns to numeric types, any rows containing NaN values in the target variables were removed using .dropna() to ensure a clean dataset for analysis.

2) **Understanding the Data:** All values in the raw American Community Survey (ACS) tables were initially stored as strings. We identified specific variables of interest, such as MedianIncome (economic indicator), PovertyRate, UnemploymentRate, and MedianAge (demographic vulnerability indicator). The verbose column descriptions were removed to prepare for numeric parsing, to convert these string-based columns into numerical data types.

3) **Basic Statistical Analysis:** ACS values are estimates and carry margins of error.

Data Integration and Similarity Analysis

- The GEO_ID (FIPS code) is the primary key used to link the three Data Profile tables (DP03, DP04, DP05).

- The data is integrated specifically for the year 2019 to align with the food access dataset window (2015–2019) and avoid pandemic-driven distortions.

4) **Data Quality Assessment:**

⚠ Data Ethics & Limitations

- **Survey estimates carry margins of error** — ACS values are estimates, not exact counts. Margin-of-error columns (`*M`, `*PM`) are dropped here but should be considered in any inference.
- **5-year pooling:** County estimates represent an average over 2015–2019 — they do not reflect a single point in time.
- **Hispanic/Latino framing:** Using ethnicity as a risk variable requires careful framing — the goal is to identify underserved populations, not to attribute food insecurity to ethnicity itself.
- **FIPS boundary changes:** Rare county mergers/splits between 2019 and the food access datasets (2010) may cause small merge losses.

5) **Advanced Data Understanding :**

The GEO_ID (FIPS code) is the primary key used to link the three Data Profile tables (DP03, DP04, DP05).Extraction of specific variables relevant to food access research, such as MedianIncome, PovertyRate, UnemploymentRate, NoVehicleRate, HousingVacancyRate, MedianAge, and HispanicShare was done.

Visualizations and Insights:

1) **Heatmap :** SocioEconomic Relationships between ACS variables
Insight : The strongest relation is between Unemployment and Structural risk and the weakest is between No Vehicle Rate vs. Mobility Barrier.

2) **Histogram :** Income Inequality across U.S Counties
Insight : Most U.S. counties have a median household income concentrated between $40,000 and $60,000. The histogram is right-tailed. This indicates that while most counties are clustered at the lower-to-middle end of the scale, there is a small number of "elite" counties with median incomes reaching $100,000 to over $140,000.

3) **Scatterplot :** Structural relationship between Unemployment and Poverty
Insight : The chart displays a clear positive correlation. As unemployment rates increase, poverty rates generally tend to rise as well.

4) **Horizontal Bar Chart :** Counties with highest structural food access risk

Insight: The bar for 02158, which corresponds to the Kusilvak Census Area in Alaska. With a structural risk of approximately 3.5, it sits significantly higher than any other area on the list.

5) **Scatterplot :** Economic and Mobility barriers to food access
   Insight : Strong positive correlation between PovertyRate and MobilityBarrier. The Outlier– A single point at roughly (3.5, -30) shows a moderate-to-high poverty rate combined with a catastrophic mobility barrier and the highest structural risk on the chart.

6) **Histogram :** Age structure across U.S Counties
   Insight : The highest frequency of counties falls within the -0.5 to 0.5 standard deviation range. The data follows a near - normal distribution. On the left side, we see a tail extending toward -3.5. These represent counties with significantly younger populations than the national average.

7) **Scatterplot :** Income vs Transportation Access
   Insight : There is an inverse relationship between Medianincome and NovehicleRate households. As the median household income increases, the percentage of households without a vehicle significantly drops.

8) **Histogram :** Transportation barriers across US Counties
   Insights : The vast majority of U.S. counties are clustered near the **0%** mark. Most counties have nearly 0% households without a vehicle. There are extreme cases where % Households Without Vehicle reaches as high as 16% to 17%. These represent isolated counties where transportation barriers are exceptionally high.

9) **Choropleth Map :** Geographic Distribution of Structural Food Access Risk
   Alaska is the most prominent outlier on the map. Large swaths of the state are deep red, indicating extreme structural risk.

10) **Choropleth Map :** .Structural Food System Risk Across U.S. Counties
    Insight : Much of the central U.S. (Iowa, Nebraska, Illinois) shows lower value of Structuralrisk_norm. Parts of the Mid-Atlantic and New England show moderate to low value of Structuralrisk_norm.

Before/After:

Before Cleaning : ACS tables include two types of uncertainty columns:
- `*M` columns — raw margin of error (e.g., `DP03_0062EM`)
- `*PM` columns — percent margin of error (e.g., `DP03_0062EPM`)

All values were stored as strings, even though the vast majority represented numeric estimates or percentages.

The raw table contains **hundreds of columns** — all ACS DP03 variables including estimates, margins of error (`*M`), and percent margins (`*PM`). Column names are census variable codes (e.g., `DP03_0062E`). All values are stored as strings.

```
acs_econ.head()
```

	GEO_ID	NAME	DP03_0001E	DP03_0001M	DP03_0002E	DP03_0002M	DP03_0003E	DP03_0003M	DP03_0004E	DP03_0004M	...	DP03_0133PM	DP03_0134PE	DP03_0134PM	DP03_
0	0500000US01001	Autauga County, Alabama	43953	259	25908	671	25458	689	24522	738	...	1.6	13.6	1.8	
1	0500000US01003	Baldwin County, Alabama	172297	329	99495	1720	99317	1737	95091	1811	...	0.8	10.2	1.0	
2	0500000US01005	Barbour County, Alabama	20636	101	9262	444	9262	444	8413	448	...	2.4	27.6	2.8	
3	0500000US01007	Bibb County, Alabama	18492	140	9046	567	9046	567	8387	615	...	3.6	17.1	4.4	

After Cleaning : Placeholders like (X) and - were replaced with NaN (Not a Number) to prevent errors during calculation.String values were converted to floats or integers to enable mathematical analysis. The separate files were merged into a single cohesive dataframe using the GEO_ID as a unique identifier.All research-critical columns are in numeric formats, allowing for immediate statistical modeling or visualization.

```
acs_clean.shape
acs_clean.head()
```

	CountyFIPS	MedianIncome	PovertyRate	UnemploymentRate	NoVehicleRate	HousingVacancyRate	MedianAge	HispanicShare	CountyName	State
0	01001	58731	12.3	34.7	4.4	8.9	38.2	76.8	Autauga County	Alabama
1	01003	58320	6.9	39.8	3.3	29.1	43.0	86.2	Baldwin County	Alabama
2	01005	32525	23.7	45.2	10.2	22.2	40.4	46.8	Barbour County	Alabama
3	01007	47542	15.0	42.9	6.2	25.0	40.9	76.8	Bibb County	Alabama
4	01009	49358	9.6	38.5	4.1	14.3	40.7	95.5	Blount County	Alabama